AI CENTRALIZED OBSERVABILITY PLATFORM

Complete Frontend Architecture, Modules, Features and Operational Flow

This document consolidates the complete frontend and platform structure discussed for an AI-powered centralized logging, monitoring, alerting, reporting, AIOps and automated remediation platform. Logging and Monitoring are independent modules that can be enabled separately per host, while the AI/AIOps layer can correlate data from both when required.

1. Platform Top-Level Structure

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AI CENTRALIZED OBSERVABILITY PLATFORM
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■■■ Overview / Main Dashboard
■■■ Host Management
■■■ Logging
■■■ Monitoring
■■■ AI / AIOps
■■■ Automation & Remediation
■■■ Incident Management
■■■ Reporting
■■■ Retention & Archive
■■■ Integrations
■■■ Administration
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2. Overview / Main Dashboard

The main dashboard provides a high-level operational view of the entire environment.

Infrastructure Summary: Total Hosts, Online, Offline, Logging Enabled, Monitoring Enabled, Both Enabled, Applications and Services.

Alert Summary: Critical, High, Medium and Low alerts, including unacknowledged, acknowledged and resolved counts.

AI Summary: AI anomalies, investigations, root causes, recommendations, pending approvals and automated actions.

Overall Health: Environment-level health for Production, QA, UAT and Development.

Quick Problems: Top active infrastructure, application and service problems.

AI Executive Summary: AI-generated summary of major issues, impact and recommended actions.

3. Host Management

Hosts

Manage all registered servers with Host ID, hostname, IP, OS, environment, location, status, agent status, logging status and monitoring status.

Add Host

Register hostname, IP address, operating system, environment, location, server group, criticality, owner and tags.

Capability Selection: Enable Logging, Enable Monitoring, or enable both.

Supported modes: Logging Only, Monitoring Only, or Logging + Monitoring.

Host Details

Show host identity and two independent entry points: a Logging workspace and a Monitoring workspace. Each can be opened independently and optionally in a new browser window.

4. Logging Module

Logging is a completely independent operational workspace focused on collecting, searching, analyzing and alerting on logs.

Logging Dashboard

Display total logs, errors, warnings, critical events, log volume, logs per second, active log sources and hosts sending logs. Include log-volume trends, error trends, warning trends, critical events, top applications, top hosts and top error messages.

Log Explorer

Provide full-text search, regex search, time-range filtering, saved searches, export, download and shareable queries. Filters should include host, environment, application, service, log type, log source and log level.

Live Logs

Provide real-time streaming with pause, resume, auto-scroll, dynamic filtering and download.

Log Sources

Manage application logs, system logs, audit logs, security logs, database logs, web server logs, Docker logs, Kubernetes logs, Windows Event Logs and custom logs.

Each log source should retain Host ID, Application, Service, Log Type, Log Path or Event Source, Parser, Format and Status.

Log Rules

Create rules based on exact messages, keywords, regular expressions, frequency, repeated patterns and other conditions.

Logging Alerts

Logging alerts are generated from log entries. Examples include exact log matches, keyword matches, regex matches, error frequency thresholds and repeated error patterns.

Alert configuration should include rule name, log source, host, application, service, condition, severity, duration, notification behavior and AI analysis.

Alert actions include acknowledge, resolve, silence, assign, escalate and investigate with AI.

AI Log Analysis

Users can select logs for AI analysis. The AI should provide summary, issue, severity, risk, probable root cause, affected systems, related logs, similar historical issues, recommended solution and confidence score.

Logging Reports

Provide Daily Log Report, Weekly Error Report, Monthly Log Summary, Application Error Report, Security Log Report, Audit Log Report, Critical Event Report and Top Error Report.

Reports should include total logs, errors, critical events, top error patterns, top applications, top hosts, log volume, AI analysis and recommended actions. Support daily, weekly, monthly or custom schedules and PDF, CSV, Excel and email delivery.

5. Monitoring Module

Monitoring is a completely independent workspace focused on infrastructure, application, service, database and metric health.

Monitoring Dashboard

Display monitored hosts, healthy, warning, critical and offline counts. Include CPU, memory, disk, network, disk I/O, load, uptime, process and availability views.

Infrastructure Monitoring

Monitor CPU, RAM, disk, disk I/O, network, load average, uptime, processes and filesystems.

Application Monitoring

Monitor application health, response time, requests per second, error rate and throughput.

Service Monitoring

Monitor Nginx, Apache, MySQL, MongoDB, Redis, Elasticsearch and custom services with Running, Stopped, Failed and Unknown states.

Database Monitoring

MySQL: Connections, queries, slow queries, CPU, memory, disk and replication.

MongoDB: Primary, secondary, replication lag, connections and operations.

Redis: Master, replica, Sentinel, memory, connections and commands.

Elasticsearch: Cluster health, nodes, JVM, disk, shards and indexing.

Monitoring Alerts

Monitoring alerts are separate from logging alerts. Examples include CPU, memory, disk, network, server offline, service down, database replication lag, high API response time and high HTTP 5xx.

Threshold Management

Users can configure warning and critical thresholds and define how long a condition must remain true before an alert is generated.

Threshold Profiles

Create reusable profiles such as Production Linux, Production Database, Production Windows, QA Linux and Development. Assign profiles to hosts, multiple hosts, host groups, applications or environments.

Monitoring Reports

Provide Daily Infrastructure Health, Weekly Infrastructure Report, Monthly Infrastructure Report, Server Availability, CPU Performance, Memory Performance, Disk Capacity, Network Performance, Application Performance, Database Health, Service Availability and SLA/Uptime reports.

Reports should include CPU and memory averages and peaks, disk growth, network traffic, downtime, availability percentage, alerts, anomalies and AI recommendations. Support daily, weekly, monthly or custom schedules and email, PDF, Excel and CSV delivery.

6. AI / AIOps

The AI/AIOps layer sits above Logging and Monitoring and analyzes logs, metrics, alerts, events, historical incidents, deployments, configuration changes and the knowledge base.

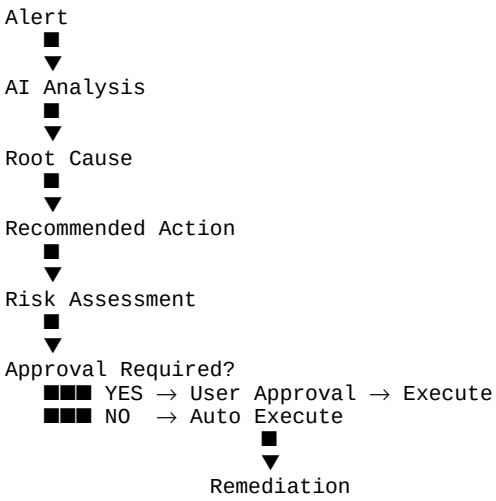
Core capabilities include AI Log Analysis, AI Metric Analysis, Anomaly Detection, Root Cause Analysis, Risk Classification, AI Recommendations and AI-generated reports.

Risk levels: LOW, MEDIUM, HIGH and CRITICAL.

AI should correlate logs and metrics when investigating root cause. For example, a CPU alert can trigger investigation of related application logs, database errors and historical events.

7. Automation & Remediation

The platform integrates with an n8n AI Agent for controlled remediation.



Possible actions include restarting services, clearing caches, restarting applications, archiving logs, cleaning temporary files, scaling services, running scripts, executing Ansible playbooks and triggering n8n workflows.

Production environments should generally use approval workflows, while lower-risk environments may allow automatic execution based on policy.

8. Incident Management

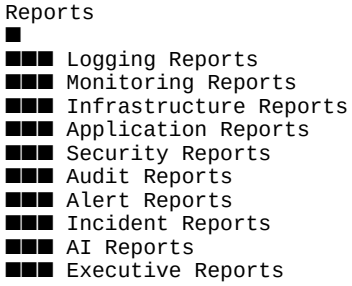
Alerts and incidents are separate concepts. An incident can combine multiple related alerts, logs and metrics into one operational problem.

Incident lifecycle: Detected, Investigating, Identified, Mitigating, Resolved and Closed.

An incident should contain title, severity, status, affected systems, related alerts, AI investigation, root cause, impact, remediation actions and resolution.

9. Central Reporting

Provide a dedicated Reports module in addition to Logging-specific and Monitoring-specific report areas.



The report scheduler should allow users to select data sources such as logs, metrics, alerts, incidents and AI analysis; define schedules; select recipients; and choose output formats.

10. Retention & Archive

Configure retention for logs, metrics, alerts, incidents and audit records. Example lifecycle: hot storage for 30 days, archive for 365 days and deletion after 730 days.

Support local storage, S3, MinIO and other object storage. Features should include archive, restore, delete, storage monitoring, archive status and retention policies.

Retention, archive and delete defaults can be controlled through environment variables, while the frontend can expose controlled policy management if dynamic changes are permitted.

11. Integrations

Recommended integrations include the existing collection agent, n8n, Ollama, cloud LLM APIs, Loki, Elasticsearch, Prometheus, Alertmanager, S3, MinIO, email, Slack, Microsoft Teams and webhooks.

12. Administration

Users and Roles

Recommended roles include Admin, DevOps, NOC, Developer, Viewer and Auditor.

RBAC

Control permissions for viewing logs, viewing metrics, creating rules, modifying thresholds, configuring hosts, approving automation, executing actions, viewing reports and managing users.

Audit Logs

Record who performed an action, what changed, when it happened, which host or resource was affected, old value, new value and result.

13. Complete Frontend Navigation

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AI CENTRALIZED OBSERVABILITY PLATFORM
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■■■ ■ OVERVIEW
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■■■ ■■ HOST MANAGEMENT
■   ■■■ Hosts
■   ■■■ Add Host
■   ■■■ Host Groups
■   ■■■ Agents
■
■■■ ■ LOGGING
■   ■■■ Logging Dashboard
■   ■■■ Hosts
■   ■■■ Applications
■   ■■■ Log Sources
■   ■■■ Live Logs
■   ■■■ Log Explorer
■   ■■■ Log Patterns
■   ■■■ Log Rules
■   ■■■ Log Alerts
■   ■■■ AI Log Analysis
■   ■■■ Logging Reports
■   ■■■ Logging Retention
■
■■■ ■ MONITORING
■   ■■■ Monitoring Dashboard
■   ■■■ Hosts
■   ■■■ Infrastructure
■   ■■■ Applications
■   ■■■ Services
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■   ■■■ Databases
■   ■■■ Metrics
■   ■■■ Thresholds
■   ■■■ Monitoring Alerts
■   ■■■ AI Monitoring Analysis
■   ■■■ Monitoring Reports
■   ■■■ Monitoring Retention
■
■■■ ■ AI / AIOPS
■   ■■■ AI Dashboard
■   ■■■ Anomaly Detection
■   ■■■ Root Cause Analysis
■   ■■■ Risk Classification
■   ■■■ Recommendations
■   ■■■ AI Reports
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■■■ ■■ AUTOMATION
■   ■■■ Pending Approvals
■   ■■■ Automation Rules
■   ■■■ n8n Workflows
■   ■■■ Executions
■   ■■■ Action History
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■■■ ■ INCIDENTS
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■■■ ■ REPORTS
■   ■■■ Logging Reports
■   ■■■ Monitoring Reports
■   ■■■ Alert Reports
■   ■■■ Incident Reports
■   ■■■ AI Reports
■   ■■■ Scheduled Reports
■
■■■ ■ RETENTION & ARCHIVE
■
■■■ ■ INTEGRATIONS
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■■■ ■ ADMINISTRATION
    ■■■ Users
    ■■■ Roles & RBAC
    ■■■ Audit Logs
    ■■■ System Settings
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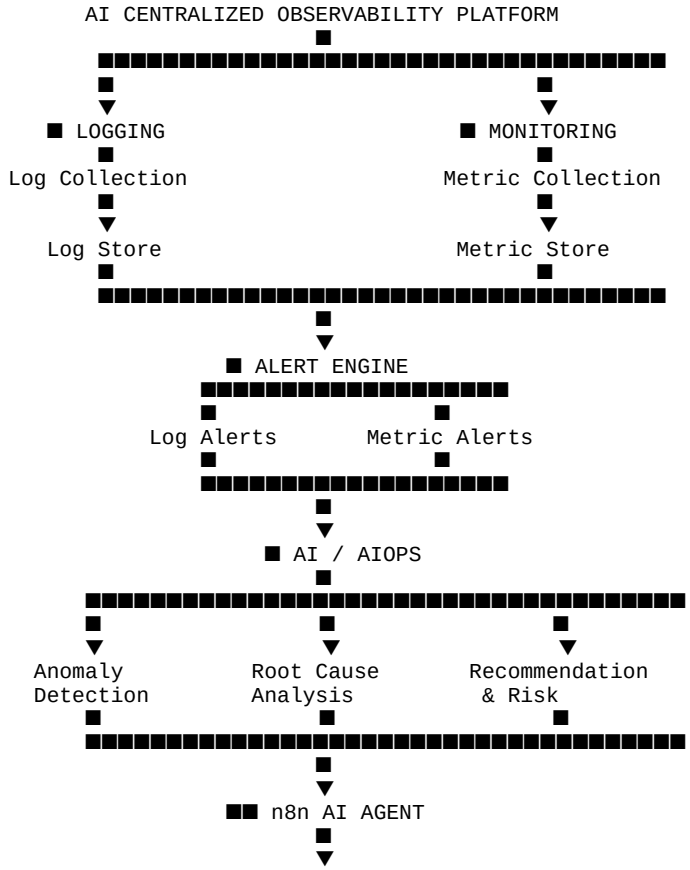
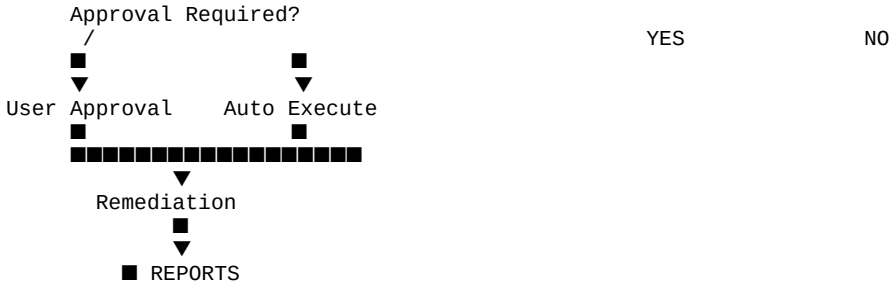
14. Complete Operational Architecture





15. Core Architecture Principle

Logging and Monitoring are independent capabilities at the host level. A host can have Logging ON and Monitoring OFF, Monitoring ON and Logging OFF, or both ON.

The frontend should present Logging and Monitoring as separate workspaces, and each workspace can optionally be opened in a separate browser window. The backend can still correlate the two through shared metadata such as host_id, environment, application, service and time.

The AI/AIOps layer sits above both modules and correlates them only when required for anomaly investigation, root cause analysis, recommendations and automated remediation.

Example end-to-end flow: Monitoring detects CPU at 98% → Monitoring alert fires → AI checks related application logs → AI discovers repeated database errors → AI identifies probable root cause → AI recommends remediation → n8n executes after approval → monitoring verifies recovery → incident is resolved → result is included in scheduled reporting.